Materials and Methods

Preparation of Spawn

Select healthy and clean wheat straw and boil in water (15-20 min.).Remove excess water on sieve**.** And dry in shade Incubate for 20-25 days (shake after 10 days) and the Spawn is ready

Spawning and Filling of bags and trays

For spawning, the mixed spawning method was used. Mixing of wheat grain-based spawn was done @ 2 kg per 100 kg compost of *Agaricus bisporus* under clean conditions (i.e., with clean hands and a pre-sterilized area). Good quality compost with a temperature of 25 °C was used. Then fill spawned compost into polythene bags (size 16” x 22”) to a depth of 10-12”. Little compressing and leveling of spawned compost was done.

Casing

A mixture was prepared and sterilized. It was added up to 3 cm on top of the spawn run compost. Sterilization was done by making a heap of casing material on a cemented platform and wetting up to 50-60% water holding capacity. The wet casing was drenched by mixing with the shovel. It was then covered with a polythene sheet and the outer periphery was sealed. The material was kept for 24-48 hrs. The cover was removed after 48 hrs.’ and the material was exposed to open air and sunlight by spreading over with clean tools and permitting to escape into air for 2-3 days before it was used as casing.

Cropping

Under favorable environmental conditions viz. temperature initially 23 ± 2 °C for about a week and then 16 ± 2 °C), moisture (2-3 light sprays per day for moistening the casing layer), humidity (above 85%), proper ventilation and CO2 concentration (0.08-0.15 %) the fruit body initials which appeared in the form of pinheads started growing and gradually developed into button stage. The first crop appeared about three weeks after casing. Mushrooms were harvested by gentle twisting and soil end parts were cut off. After harvesting was complete, the gaps in the beds were filled with fresh sterilized casing material and then watered. Yield data for average weight and total yield of fruiting bodies per bag was recorded

Crude protein extraction

SDS-PAGE Electrophoresis

Primary Sequence Analysis of AbL

AbL sequence was retrieved from the NCBI database https://www.ncbi.nlm.nih.gov with NCBI Accession No. ACC18877. Translated protein sequence from this gene was also retrieved from this gene sequence by NCBI Accession No.: ACC18877 This enzyme sequence can also be retrieved from the UniProtKB database (https://www.uniprot.org) by (UniProtKB ID: Q12541).

Multiple Sequence Alignment and Sequence Homology Analysis

After retrieval of the FASTA sequence of AbL, BLAST-P was run on the NCBI website with PDB selected as the database. A homology modeling template search was done on the SWISS-MODEL to get the most suitable template for AbL. Possible templates for the AbL were selected depending on different values, and multiple sequence alignment (MSA) was done on Clustal Omega, https://www.ebi.ac.uk/Tools/msa/clustalo/, which is a multiple sequence alignment program that produces biologically meaningful multiple sequence alignments of divergent sequences. A percentage Identity matrix homology table among those selected templates was built by this alignment.

Multiple sequence alignment was done on the ESPript 3 server for proper understating of the homology sequence. The alignment file was downloaded from the Clustal Omega server, and the predicted model of AbL was uploaded to the ESPript 3 server to get the MSA (https://espript.ibcp.fr/ESPript/ESPript/index.php). It is a tool of choice for biologists, and structural biologists, enabling production with a few mouse clicks, a collection of comprehensive, high-quality figures, and interactive 3D representations of their proteins of interest. With its simplified interface and capacity to rapidly generate the significant properties of any protein structure, it is also an efficient instructional tool that may be used during courses or practical work sessions in structural biology among these templates (Robert & Gouet, 2014).

Phylogenetic Tree

A phylogenetic tree among all the possible selected templates was constructed. An accurate phylogenetic tree is essential for inferring the genesis of novel genes, identifying molecular adaptation, comprehending morphological feature evolution, and recreating demographic changes in recently separated species. The Neighbor-Joining approach was used to infer the evolutionary history (Saitou & Nei, 1987). The Poisson correction technique was used to calculate the evolutionary distances in the units of the number of amino acid substitutions per site (Zuckerkandl & Pauling, 1965). This study comprised five amino acid sequences. All ambiguous sites were deleted for each sequence pair (pairwise deletion option).

Three Dimensional Structure Prediction of AbL

FASTA sequence of AbL was uploaded on the SWISS-MODEL (https://swissmodel.expasy.org/) online server (Waterhouse et al., 2018). It is a web-based integrated service focused on protein structure homology modeling. The SWISS-MODEL database includes automatically produced homology models for appropriate model organisms and quantitative structural information for all UniProtKB sequences. It aids the user in generating protein homology models at various degrees of complexity. Structure building through this server includes the five steps: (i) Input data (ii) Template search (iii) Template selection (iv) Model building (v) Model quality estimation.

The predicted structure was downloaded in PDB format, and all its essential sites were highlighted and labeled along with its domains in PyMOL software (The PyMOL Molecular Graphics System, Version 2.5.2 Schrödinger, and LLC). The predicted model of AbL was uploaded on the PDBsum. This web server gives a summary and visual assessment of empirically verified macromolecular structure models in the Protein Data Bank (PDB). In addition, the PDBsum server enables you to submit your PDB-format data and produce and file a whole collection of PDBsum structural analyses and schematics diagrams depicting the interactions (Laskowski, 2001). A 2D structure was generated and labeled with helices, beta sheets, loops, all crucial residues, and domains.

Structure Refinement and Validation

The predicted model of AbL was refined on the Galaxy refine webserver (https://galaxy.seoklab.org/). A significant characteristic of the server from existing protein

structure servers is that unreliable sections for which template information is not accessible or inconsistent are recognized and refined using an *initio* process (Ko *et al*., 2012). This refined model can be used for the next step of docking after validation.

The generated model from SWISS-MODEL was validated at the Molecular Biology Institute and the DOE-MBI Institute online servers at the University of California, Los Angeles (https://saves.mbi.ucla.edu/) online server. The structure was validated in many ways, like ERRAT, Ramachandran plot, VERIFY 3D, etc.

ERRAT is based on the concept that various atom types will be distributed non-randomly in proteins owing to complicated geometric and energetic considerations. Structural flaws would result in visible abnormalities in the pattern of interactions. The following constraints apply to the evaluation of non-bonded interactions: the gap in space between two atoms is smaller than some predetermined limit, often 3.5 Å, and atoms within the same residue or those covalently bound to one another are not regarded. This distribution is used to assess the likelihood that a particular collection of interactions from a protein model is accurate. Because the ERRAT assessment is based on a standard distribution calibrated on a credible database, estimating the chance that each area of a candidate protein model is wrong is simple. This approach identifies erroneously generated areas in protein models in an impartial and statistically sound manner (Colovos & Yeates, 1993).

The PROCHECK suite of programs checks the stereochemistry of a protein structure in detail. Its outputs include a variety of PostScript graphics and a complete residue-by-residue listing. These provide an evaluation of the model's overall quality compared to well-refined structures of the same resolution, as well as highlighting locations that may need more examination (Laskowski *et al.*, 1993).

The Ramachandran plot is critical in identifying the stereochemical quality of the protein model. The conformation angles, also known as the Ramachandran angles, dictate the backbone of a protein molecule's polypeptide chain. In the Ramachandran Plot, these angles are plotted against one other. The Ramachandran plot is a powerful tool for analyzing the conformation angles of any three-dimensional structure determined using X-ray crystallography or NMR. Users may upload the three-dimensional atomic coordinates of a protein molecule of interest from the client machine (Gopalakrishnan *et al*., 2007).

The compatibility of a 3D protein model to its amino acid sequence, as determined by a 3D profile, is an effective test of the model's accuracy. A protein structure's 3D profile is a table generated from the structure's atomic coordinates that may be used to score the compatibility of the three-dimensional structure model with any amino acid sequence. High-scoring three-dimensional profiles generated from proper protein structures match their sequences. In contrast, 3D profiles computed from incorrect protein models perform poorly. Using a 3D profile, the approach assesses the compatibility of a protein model with its sequence. Each residue location in the 3D model is defined by its environment and is represented in the profile by a row of 20 integers. These are the statistical preferences (referred to as 3D-1D scores) of each of the 20 amino acids in this environment (Eisenberg *et al*., 1997).

Structural Alignment of AbL and its template

After selecting the template, the alignment between the template and target sequence was constructed in the PyMOL program and their RMSD values were interpreted. The structural differences between two ideally aligned models are commonly assessed as the Root Mean Square Deviation (RMSD) between the aligned alpha-carbon locations (excluding deviations from the non-aligned positions). To offer a frame of reference for RMSD measurements, up to 0.5 Å RMSD of alpha carbons occurs in separate determinations of the same protein. An optimum RMSD value is always less than 1 Å.

Phytochemical compounds Selection

The phytochemical compounds database, having many molecules was selected for virtual screening against AbL to find potential inhibitors to stop its functioning. These compounds database was chosen from the PubChem Classification Browser (well-known and reliable chemical compound .https://pubchem.ncbi.nlm.nih.gov/classification/#hid=5.

Molecular Docking of AbL with Glycerol

Molecular docking of AbL was done with the natural substrate Glycerol (GLU) to get the residues interacting with AbL active site. The binding pocket for this docking, and the docking score was also observed. Virtual screening of small ligand molecules helps to find potential inhibitors against proteins or enzymes by site-directed target-based docking. The

MOE (Molecular Orbiting Environment) module has been used in a lot of research to carry out molecular docking of different ligands against several biological targets. Molecular docking of the selected phytochemicals was done in the MOE (v2015) https://www.chemcomp.com/.

Preparation of Protein Structure

The protein structure was prepared in module by uploading the predicted protein structure in PDB format. Using Protonate 3D, the hydrogen bond networks were re-optimized. In the context of an all-atom model of a macromolecular structure, the Protonate 3D program assigns protonation states from a discrete collection of states by minimizing the titration free energy of all groups (including ligands and solvents).

Preparing Ligand Database

All ligands downloaded from were uploaded in SDF file format; all of them were energy minimized by default parameters, and the ligands database was saved All ligands are typically utilized in their pre-reacted state. If necessary, manual structure alterations are made otherwise, they are used directly from their source database.

Analysis of AbL

Docking of the natural substrate, GLU, was done with AbL, and its interacting residues forming hydrogen bonds were used as active sites of AbL Site finder feature is available which tells us about the binding pockets present in the protein or enzyme. The site finder indicated binding pockets; a search was done for which pocket had residues interacting with the natural substrate. So, Site 1 was found suitable, and dummy atoms were created at the selected binding pocket.

Docking Analysis

Docking results obtained were analyzed in PyMOL, which was used for the generation of protein and ligand cartoon models, and LigPlot software was used for the development of LigPlot images and analysis of molecular interactions of different complexes.